

Name (Print): _____

This exam contains 11 pages (including this cover page) and 9 problems. Check to see if any pages are missing. Print your name above.

No notes will be allowed on the exam. A calculator is permitted on the exam.

Read the following instructions carefully:

- You can only use methods discussed in class so far. i.e., you may **NOT** use any method we haven't learned in class so far.
- **Show ALL your work!** You are being assessed on your work and explanations. Correct answers with little to no work or explanation will not receive full credit. I give partial credits so don't leave a blank if you have some ideas.
- **Organize your work.** Your work should be clearly written and organized. Please use correct mathematical notations when appropriate. For example, please use vector notations when working with vectors and don't use vector notations when working with scalars and points.
- By submitting this exam, you agree that all work represents your individual effort and understanding. Failure to comply with any of the above statements will be considered a violation of the Student Code of Conduct and will result in a 0 for this assessment and possible disciplinary action.

Problem	Points	Score
1	12	
2	12	
3	14	
4	12	
5	12	
6	12	
7	14	
8	12	
9	0	
Total:	100	

Do not write in the table to the right. Adding the bonus, the total score won't exceed 100%.

Please keep in mind that an assessment is just another means of communication between you and me to help me understand what you have learned so far. It is not a measure of your worth as a person nor your intelligence or aptitude as a student. Just give this your best try.

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1. (12 points) Solve the following system of linear equations using the Gaussian Elimination algorithm.

$$\begin{cases} x - 3y + 2z = 5 \\ 2x + y - z = -1 \\ 3x - 2y + z = 8 \end{cases}$$

If the system is consistent, write the solution set as a column vector. If the system is inconsistent, explain how you know it.

2. (12 points) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation such that

$$T\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} \quad \text{and} \quad T\left(\begin{bmatrix} 5 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$$

Find $T\left(\begin{bmatrix} -4 \\ -3 \end{bmatrix}\right)$.

3. (14 points) Let $A = \begin{bmatrix} 1 & 0 & 2 & -1 & 3 \\ 2 & 1 & 1 & 0 & 5 \\ -1 & 3 & -11 & 7 & -6 \end{bmatrix}$.

(a) Find a basis for $\text{Col}A$. Then determine its dimension.

(b) Find a basis for $\text{Nul}A$. Then determine its dimension.

4. (12 points) Let A , B , and C be $n \times n$ matrices such that

$$\det A = 4, \quad \det B = 5, \quad \text{and} \quad \det C = -2$$

Compute the value of $\det (A^{-1}B^3C^TA^2B^{-2})$.

5. (12 points) Let $A = \begin{bmatrix} 2 & 3 & 1 & -1 \\ 0 & 0 & 4 & 0 \\ 5 & 0 & 1 & -2 \\ 1 & -1 & 0 & 3 \end{bmatrix}$ be a 4×4 matrix.

(a) Compute $\det A$.

(b) Is A an invertible matrix based on your answer in part (a)? Justify your reasoning.

6. (12 points) Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & -1 & 2 \end{bmatrix}$ be a 3×3 matrix.

(a) Find *all* the eigenvalues of A .

(b) Is A an invertible matrix based on your answer in part (a)? Justify your reasoning.

7. (14 points) Let $A = \begin{bmatrix} 5 & 0 & -3 \\ 3 & 2 & -3 \\ 6 & 0 & -4 \end{bmatrix}$ be a 3×3 matrix with the eigenvalues of $\lambda = 2$ (with multiplicity 2) and $\lambda = -1$ (with multiplicity 1). Determine if A is diagonalizable. If so, find the matrices P and D . If not, explain why.

8. (12 points) Determine if the following statement is true or false. If it is true, include a brief proof; if it is false, provide a counterexample.

(a) Every vector in an eigenspace of a matrix is an eigenvector of the matrix.

(b) If $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent eigenvectors, then they must correspond to distinct eigenvalues.

9. (10 points (bonus)) **This is a BONUS problem.**

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation defined by $T(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & -6 \end{bmatrix}$$

We define the **image space** of a linear transformation as the set of all possible output vectors. That is,

$$\text{Im}(T) = \{T(\mathbf{x}) \mid \mathbf{x} \in \mathbb{R}^3\}$$

We define the **kernel** of a linear transformation as the set of vectors that produces $\mathbf{0}$ as an output. That is,

$$\text{Ker}(T) = \{\mathbf{x} \in \mathbb{R}^3 \mid T(\mathbf{x}) = \mathbf{0}\}$$

Find a basis for **image space** and a basis for **kernel** of the linear transformation T .